

PRESENTATION SCIENTIFIC COMPUTING

LINMA2710

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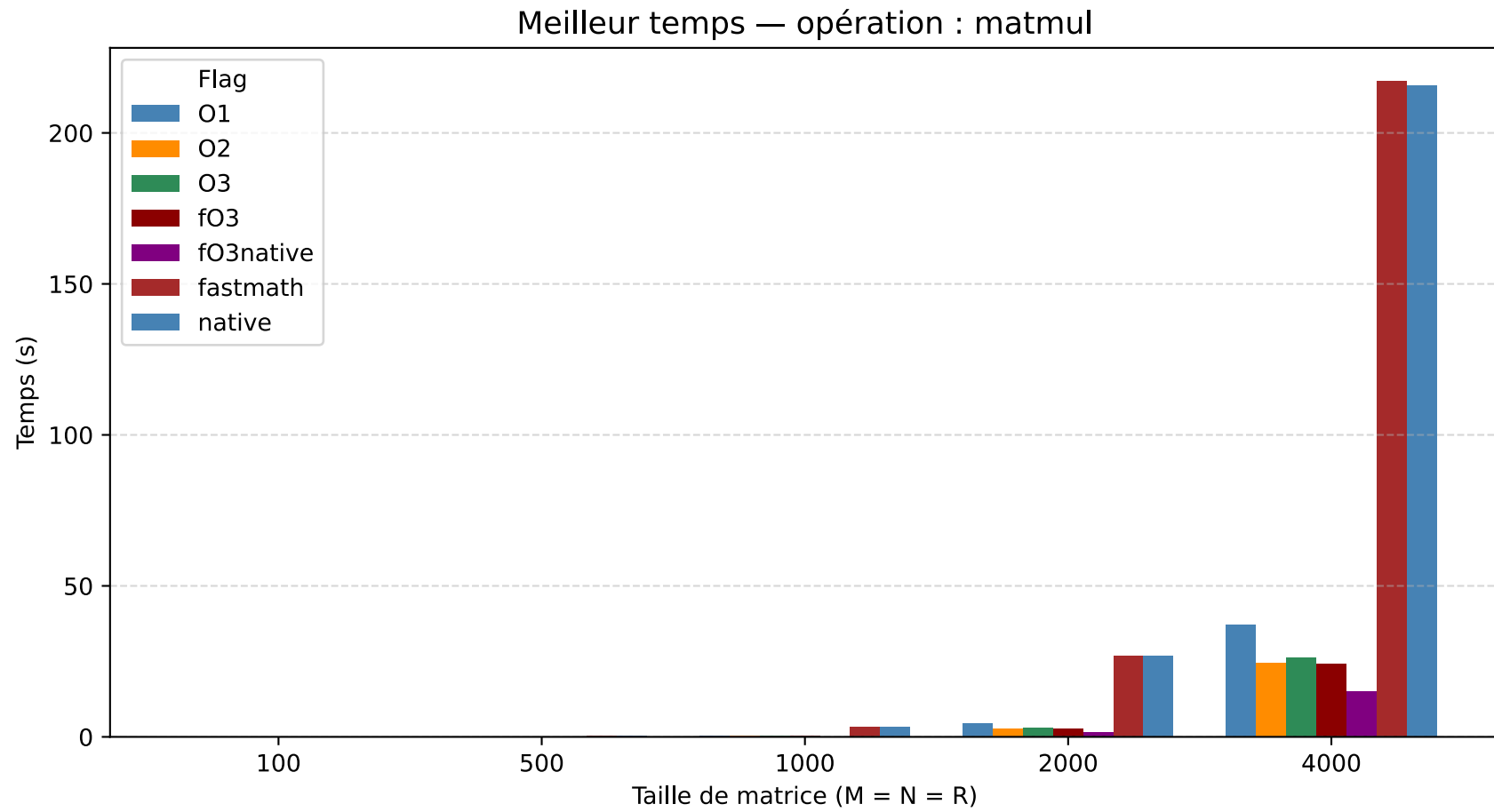
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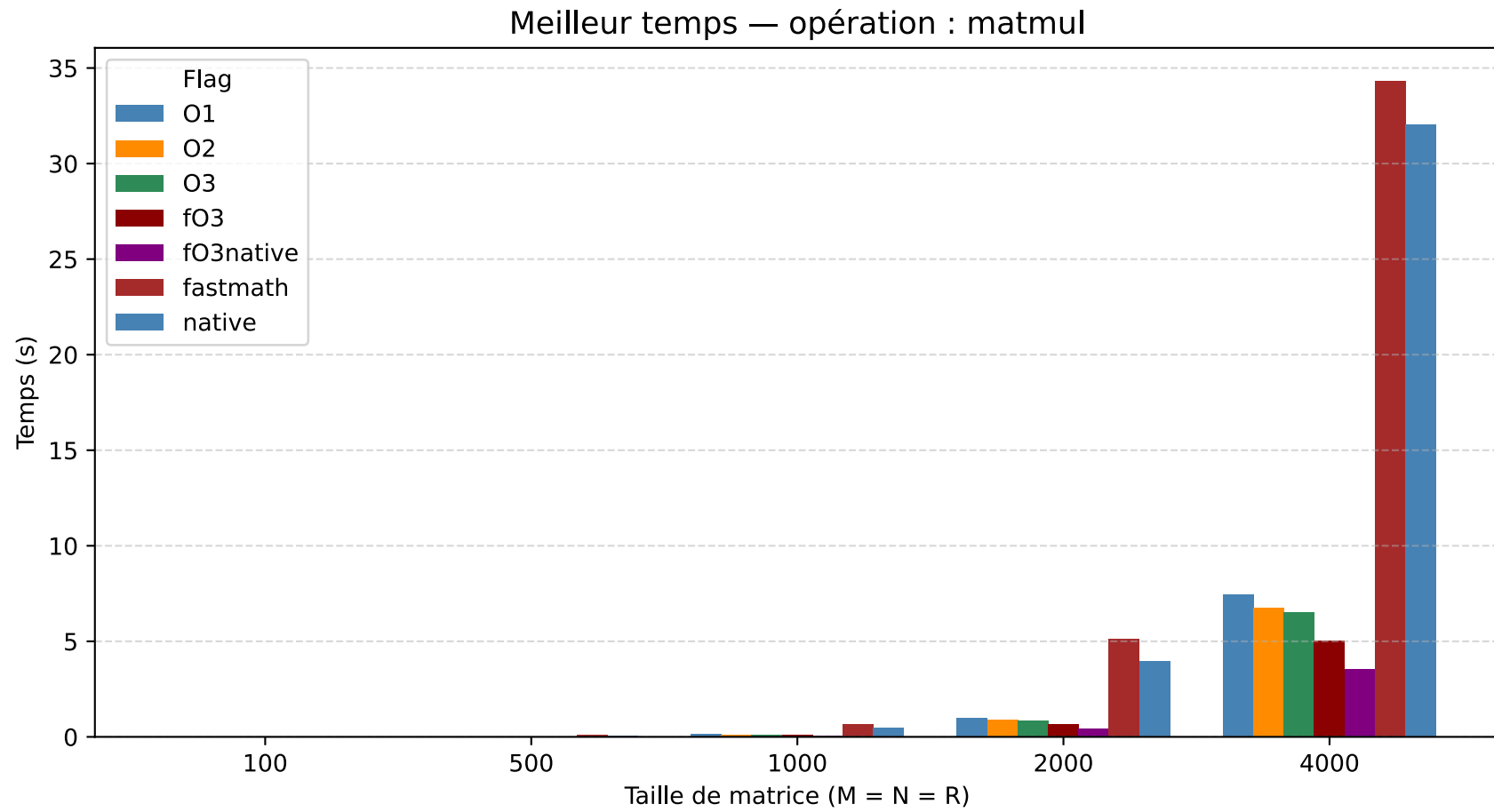
1. Part 1 : Basic matrix operation and SIMD
2. Part 2 : OpenMP
3. Part 3 : Distributed matrix operations (MPI)
4. Part 4 : GPU Matrix Operations (OpenCL)
 - Outlined slide

PART 1 : BASIC MATRIX OPERATION AND SIMD

- As the matrix is stored in row major we transpose to optimise for cach miss.
- We also do tilling to have a more efficient memory access (see later on)

- Visualisation of the time taken for each operation with each flags :
- Computation made on the cluster

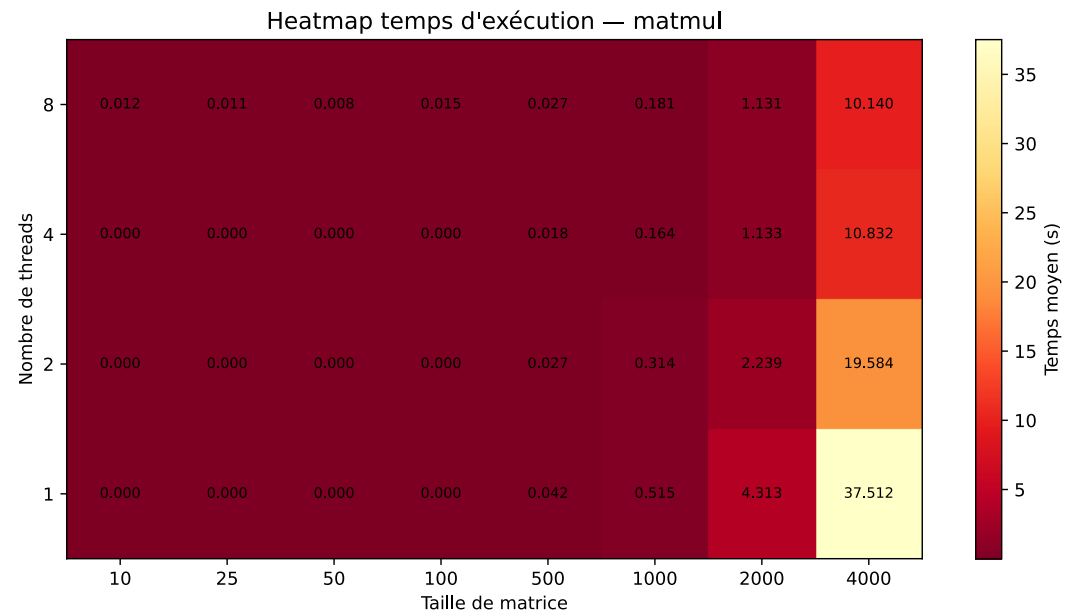




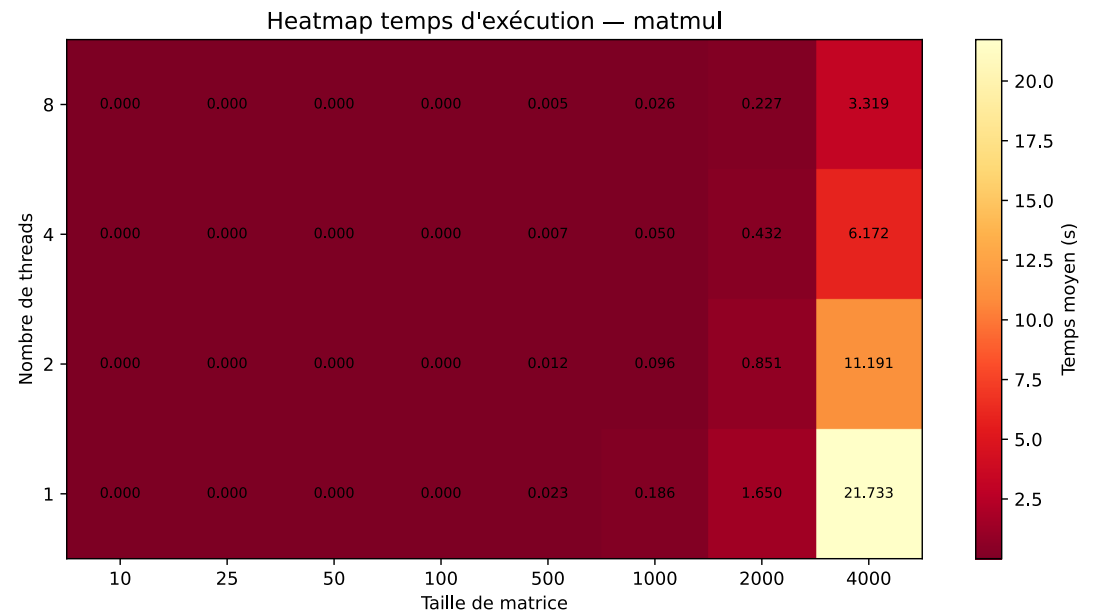
PART 2 : OPENMP

Plot of computation time VS size of matrix
VS number of threads **sans tilling** on my
pc :

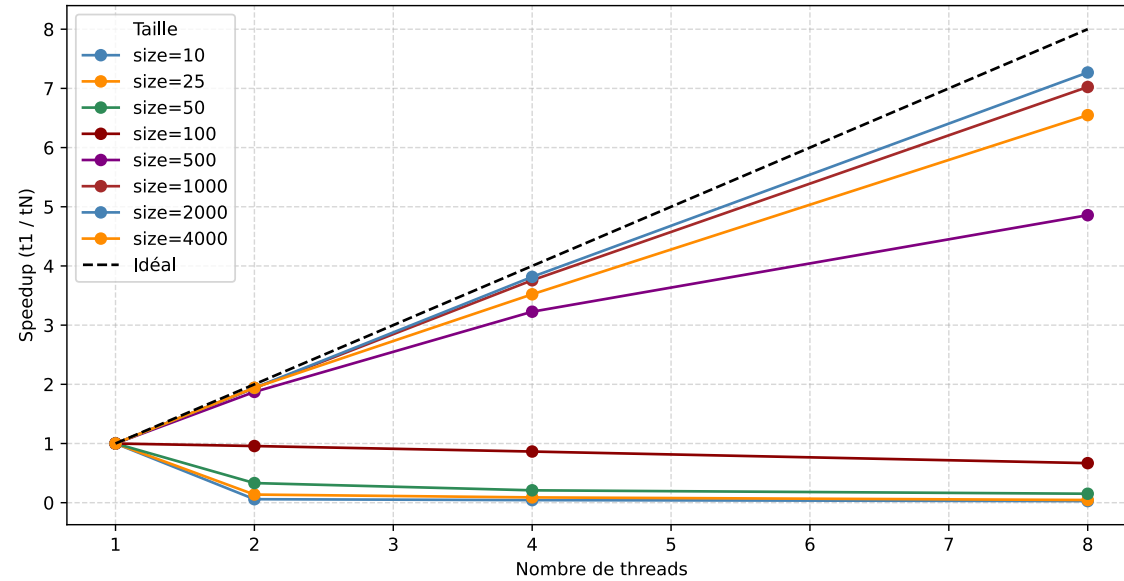
- For small matrix time to start the thread is important
- Limiting factor from 4 to 8 threads.



Plot of computation time VS size of matrix
VS number of threads **sans tilling** on the
server.



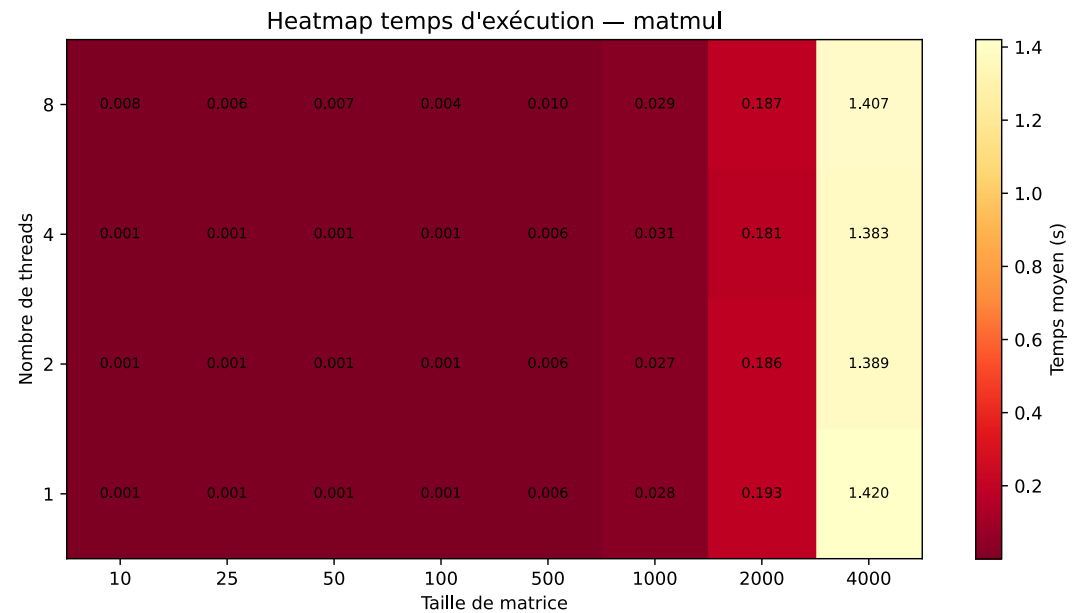
Speedup — opération : matmul



Plot of computation time VS size of matrix

VS number of threads **avec tiling** :

- Model roofline, limitation of the bandwidth



PART 3 : DISTRIBUTED MATRIX OPERATIONS (MPI)

PART 4 : GPU MATRIX OPERATIONS (OPENCL)

- This is a simple slide with no title.
- **Bold and coloured** text by using `#stress(text)`.
- Sample link: typst.app.
 - Link styling using `link-style: "color", "underline", "both"`
- Font selection using `font: "Fira Sans", size: 21pt`.

This text has been written using `#framed(text)`. The background color of the box is customisable.

Frame with title

This text has been written using `#framed(title: "Frame with title")[text]`.

This is an auto-resized *focus*
slide.

- This is a #blank-slide.
- Available **themes**¹:

bluey, reddy, greeny, yelly, purply, dusky, darky.

- Or just use **your own theme color**:
 - theme: `rgb("30500B")`

¹Use them as **color** functions! e.g., `#reddy("your text")`

- Check out the **progress bar** at the bottom of the slide.

show-progress: true

- Outline slides with outlined: true.

This is a #grayed text. Useful for equations.

$$P_t = \alpha - \frac{1}{\sqrt{x} + f(y)}$$

Columns can be
included using
`#cols[...][...]`

And this
is

an example.

- Custom spacing: `#cols(columns: (2fr, 1fr, 2fr), gutter: 2em)[...]`
 - Sample references: .
 - Add a [bibliography slide](#)...
1. `#let bib = bibliography("you_bibliography_file.bib")`
 2. `#bibliography-slide(bib)`